

Guru Nanak Dev Engineering College, Ludhiana				
Department of Applied Sciences				
Program	B.Tech.1 st Year	Semester	1	
Course Code	ESC103	Course Title	Programming for Problem Solving	
Mid Semester Examination (MSE) No.	1	Course Coordinator(s)	Pf. Jaswant Singh Taur Pf. Kuljit Kaur Pf. Meetali Pf. Harminder Kaur Pf. Maninder Kaur Pf. Kamaljeet Kaur Pf. Ranjodh Kaur Dr Sidharath Jain Pf. Palak Sood Dr. Randeep Kaur Dr. Palwinder Kaur Pf. Kawaljeet Kaur Pf. Jagdeep Kaur Pf. Nancy Pf. Amanpreet kaur Pf. Neha Pf. Ramandeep Kaur Malhi Pf. Harpreet Kaur	
Max. Marks	24	Time Duration	1 hour 30 minutes	
Date of MSE	24-09-2025	Roll Number		
Note: Attempt all questions				
Q. No.	Question		COs, RBT level	Marks
Q1	Differentiate between Syntax Error and Semantic Error with suitable example for each		CO3, L3	2
Q2	What will be the output for the following code snippet, also explain step-by-step execution? int a = 5, b = 2; float c = a / b; cout << c << endl;		CO2, L5	2
Q3	Explain the complete process that occurs behind the scenes when a C++ source code is executed to produce output. Describe each stage of compilation and execution, and illustrate the flow using a neat diagram		CO2, L3	4
Q4	Draw and label a neat block diagram illustrating the main components of a computer system and explain the function of each component		CO2, L3	4
Q5	Write a program to swap two numbers without using a temporary variable		CO2, L4	4
Q6	A company wants to analyze the salaries of its employees to understand the total payroll, average salary, and the contribution of each employee to the total salary. Write a C++ program to read the salaries of five employees, calculate the total salary, average salary, and percentage contribution of each employee to the total payroll. Also, draw a flowchart and write pseudo code for your program		CO1, CO2, L6	8
Course Outcomes (CO) Students will be able to				
1	Design simple algorithms and flowcharts for problem solving.			
2	Demonstrate modular programs involving input output operations, decision making and looping constructs by choosing the appropriate data types for writing programs.			
3	Test and execute programs and correct syntax and logical errors.			
4	Implement conditional branching, iteration and recursion.			
5	Apply the concept of arrays and string handling in problem solving.			
6	Use concept of structures and union in problem solving.			

RBT Classification	Lower Order Thinking Levels (LOTS)			Higher Order Thinking Levels (HOTS)		
RBT Level Number	L1	L2	L3	L4	L5	L6
RBT Level Name	Remembering	Understanding	Applying	Analyzing	Evaluating	Creating

